Question Number etc. in left margin | Mark allocation in right margin

Q1
a)
i) P = probability that a single frame is in error + discussion

ii) The use of a timing diagram is accepted to estimate the results

$W > 2a+1$ the value of $k \sim 2a+1$

$W < 2a+1$ the value of $k \sim W$

b) i) Edge router
- per flow traffic management
- marking packets

ii) discuss mechanism: SLA, policing, call Adm.

iii) Core router
- per class traffic management
- queueing management and scheduling based on marking at edge
- Per hop behaviour

iv) discuss mechanism: SLA, policing, call Ad.

v)

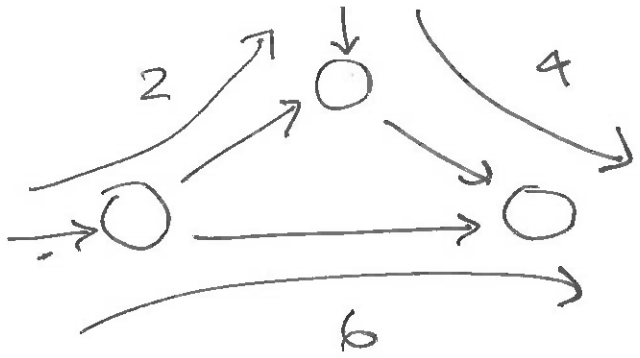
| | IntServ | DiffServ |
|---|---|---|
| - Coordination of service diff. | End-to-End | Local (per hop) |
| - Scope of service diff | unicast/ multicast | Anywhere |
| - Scalability | Limited by the number of flows | Limited by the number of classes |

Question Number etc. in left margin | Mark allocation in right margin

Q2

a)

i)




optimality condition

$$\frac{10}{(10-6)^2} > \frac{10}{(10-2)^2} + \frac{10}{(10-4)^2}$$

so there is overflow from C(1,3) to links C(1,2) and C(2,3)

ii) To derive capacity

$$\frac{C(1,3)}{(C(1,3)-6)^2} = \frac{10}{8^2} + \frac{10}{6^2} = 0.156 + 0.277$$

$$C = 0.434\,(C-6)^2$$

$$C = 0.434\,(C^2 - 12C + 36)$$

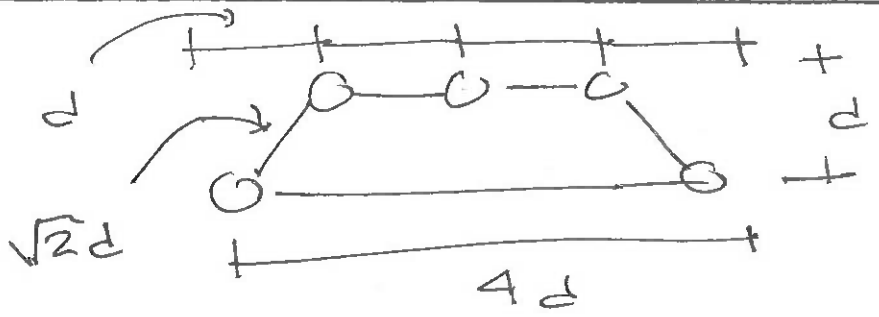
$$2.3C = C^2 - 12C + 36$$

solve: $C^2 - 14.3C + 36$

$$C \begin{cases} 3.26 \\ \boxed{11.04} \end{cases} *$$

Question Number etc. in left margin | Mark allocation in right margin

Q2 b)



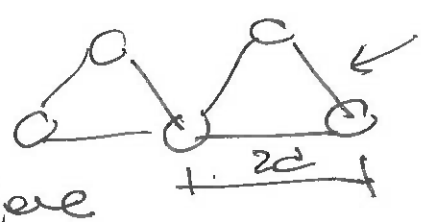

i) compare the two path in term of $P_d R^3$

$(4d)^3$ (?) $2d^3 + 2(\sqrt{2}d)^3$

$4^3$ (?) $2(2)^3 + 2(2)^{3/2}$

$4^3 < 4^3 + 2 \cdot 2^{3/2}$

ii)




Compare

$2(2d)^3$ (?) $4(\sqrt{2}d)^3$

$16d^3 > 8\sqrt{2}d^3$

iii)



discuss in term of the power consumption of the network

Question Number etc. in left margin | Mark allocation in right margin

Q3
a) i) The cost of a link and/or path cannot be expressed as a scalar. The cost could be represented by a probability distribution function or some of its moments

ii) I) for mean + variance a deterministic shortest path algorithm can be used as mean and variance of a Gaussian distribution are additive

II) mean and standard deviation cannot be solved using deterministic shortest path (standard deviation is not additive)

b) i) If all packets are of the same size round-robin schedule will be max-min fair

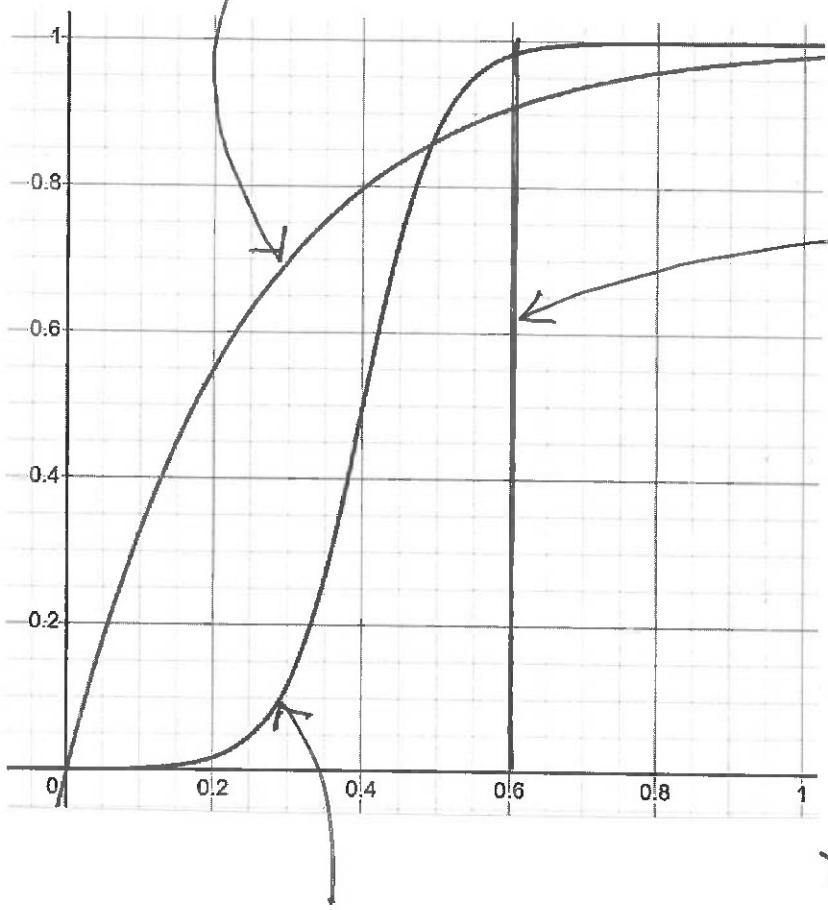
ii) The value will be $k/n$. And hence $k-n$ users will receive nothing.

This index in the most unfair scenario will be $1/n$

Question Number etc. in left margin | Mark allocation in right margin

iii)

$u_1(x) = 1 - e^{-4x}$

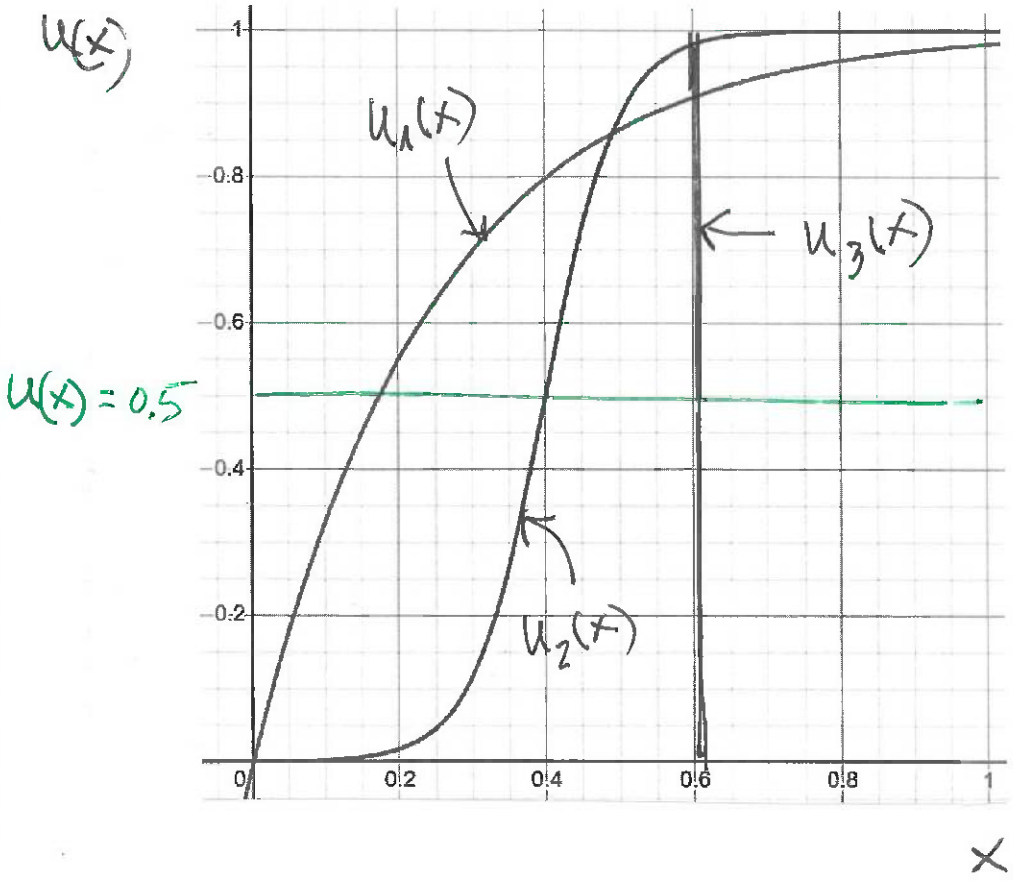



$u_3(x) = u(x-0.6)$

$u_2(x) = 1 - \frac{1}{e^{20(x-0.4)} + 1}$

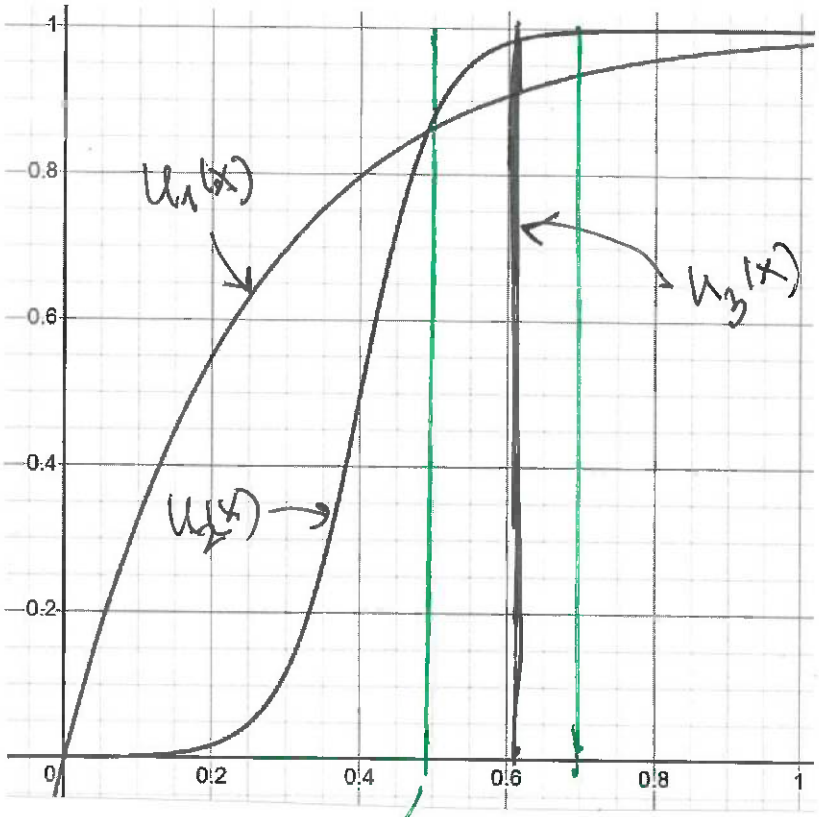
Question Number etc. in left margin | Mark allocation in right margin

iv)




discussion wrt the amount of resources x
$U_1(x)$, $U_2(x)$ and $U_3(x)$ is allocated
discussion on to the impact of this
allocation on User 3

c)




discussion wrt amount of resources x and $u_i(x)$ assigned to all utc

discussion on the impact of them allocated to utc 3 ( $u_3(x)$ ).

Question Number etc. in left margin | Mark allocation in right margin

Q4 a) i) - Lost event detected by triple duplicated ACK (in SS or CA) Threshold = cwnd / 2
Congestion window = Threshold.

- Timeout (in SS or CA) Threshold = cwnd / 2
Congestion window = 1

- discussion

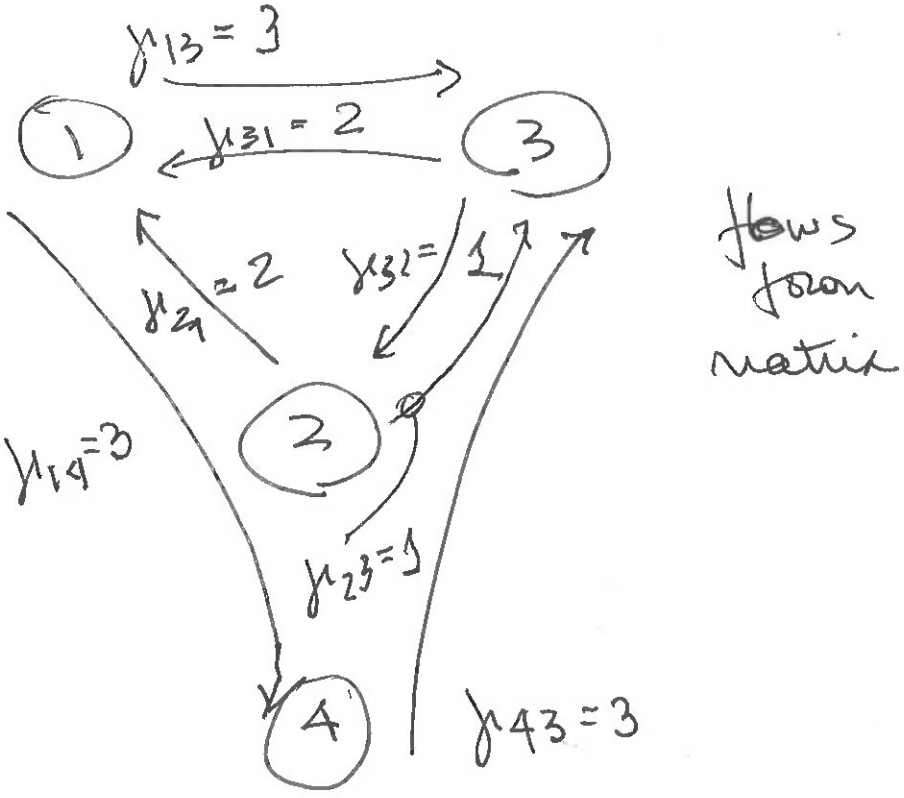
ii) Timeout interval = estimated RTT + 4 Dev RTT

RTT = Round trip time

Dev RTT = (1-μ) Dev RTT + μ (Sample RTT - estimated RTT)

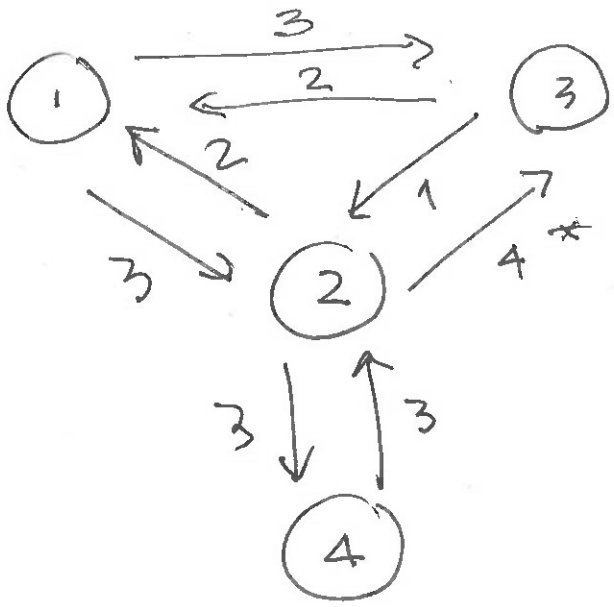
- discussion

b)



Question Number etc. in left margin | Mark allocation in right margin

i)




flow in each link

link (2,3) is the bottleneck link

ii) $$N = 4\left(\frac{3}{5-3}\right) + 2\left(\frac{2}{5-2}\right) + \frac{1}{5-1} + \frac{4}{5-4}$$

iii) Two choices need discussion
link C(3,2) and link (3,1)

link C(3,2) is carrying the lowest flow but if taken out of service will overload C(1,2)

similarly if C(3,2) is taken out of service it will overload C(2,1)

- discussion